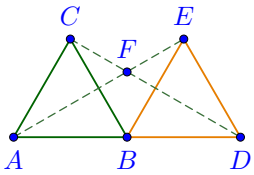
MCC Entrance Test, August 2026 — Day 1

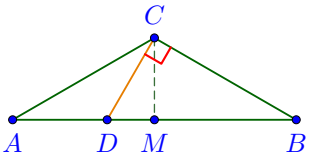
Level T3

Test Information. Level T3 — 150 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Geometry (×2), Combinatorics, Number Theory.
Rules: show your work (partial credit rewards genuine progress, but claims still need real justification at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts). Two positive integers x and y satisfy $xy + x + y = 71$ and $x^2y + xy^2 = 880$. Find $x^2 + y^2$.

Problem 2 (10 pts). Two congruent equilateral triangles ABC and BDE are positioned so that A, B, D lie on a single line, in that order, with C and E on the same side of this line. Let F be the intersection point of segments CD and AE. Find $\angle AFD$.





Problem 3 (15 pts). Triangle ABC has $AC = BC$, $AB = 3$. Point D lies on side AB so that $BD = 2\,AD$. $\angle BCD = 90°$, find $\angle BAC$.

(a) M is the midpoint of AB. Prove that $CM^2 = DM \cdot MB$. [9 pts]

(b) Using part (a), find $\angle BAC$. [6 pts]

Problem 4 (15 pts). Let a and b be real numbers with $a > b$, satisfying

$$2^a + 2^b = 75 \qquad \text{and} \qquad 2^{-a} + 2^{-b} = \tfrac{1}{12}.$$

(a) Let $x = 2^a$ and $y = 2^b$. Find xy. [6 pts]

(b) Solve for x and y, then compute 2^{a-b}. [9 pts]

Problem 5 (25 pts). Nine tokens are arranged in a row: three Copper tokens, three Iron tokens, and three Silver tokens (tokens of the same type are indistinguishable from one another). Call an arrangement *bad* if some three *consecutive* tokens are all the same type.

(a) How many distinct arrangements of the nine tokens are there, with no restriction? [4 pts]

(b) How many of those arrangements have all three Copper tokens consecutive (forming a single block), regardless of how the rest are arranged? [5 pts]

(c) Find the number of arrangements that are *not* bad. [16 pts]

Problem 6 (25 pts). A positive integer n is such that when the multiplier 2013 is applied – that is, the product $2013n$ is computed – the last four digits of the result are 2025. Find the sum of the digits of the *smallest* such positive integer n.

(a) Show that $2013n \equiv 2025 \pmod{10000}$, and find the last (units) digit of n. [8 pts]

(b) Find the tens digit of n. [5 pts]

(c) Continue this process to find the smallest positive integer n, then compute the sum of its digits. [12 pts]